# Windows 11 Networking and Sharing

Modern connectivity and file sharing made simple



# Network Visibility and Discovery

# Enabling Network Discovery



**Purpose of Network Discovery**

Network Discovery allows a PC to detect and interact with other devices on the same network, enabling seamless connectivity.

**Facilitates Sharing and Collaboration**

Enabling this feature helps with file sharing, printer access, and supports collaborative work environments effectively.

**Configuration Location**

Network Discovery can be turned on in the advanced sharing settings within the Network and Sharing Center in Windows 11.

**Importance in Trusted Networks**

Without Network Discovery, devices remain invisible to each other, reducing productivity in home and office networks.

# Choosing the Right Network Profile



**Private Network Profile**

Private networks enable device discoverability and resource sharing in trusted environments like home or work.



**Public Network Profile**

Public networks restrict discovery and sharing to protect devices in untrusted or less secure environments.

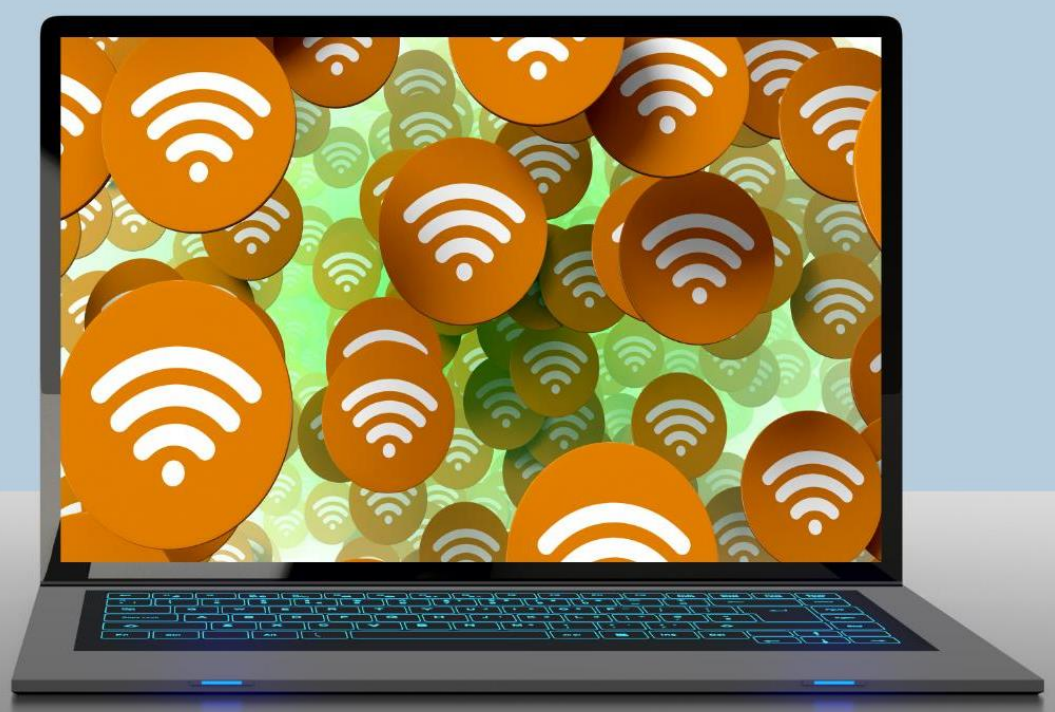


**Network Profile Importance**

Choosing the correct network profile balances security and functionality based on network trust levels.

# Connecting and Managing Network Devices

# Connecting to Wireless Networks



## Accessing Wi-Fi Networks

Click the Wi-Fi icon in the taskbar to reveal a list of available wireless networks for connection.

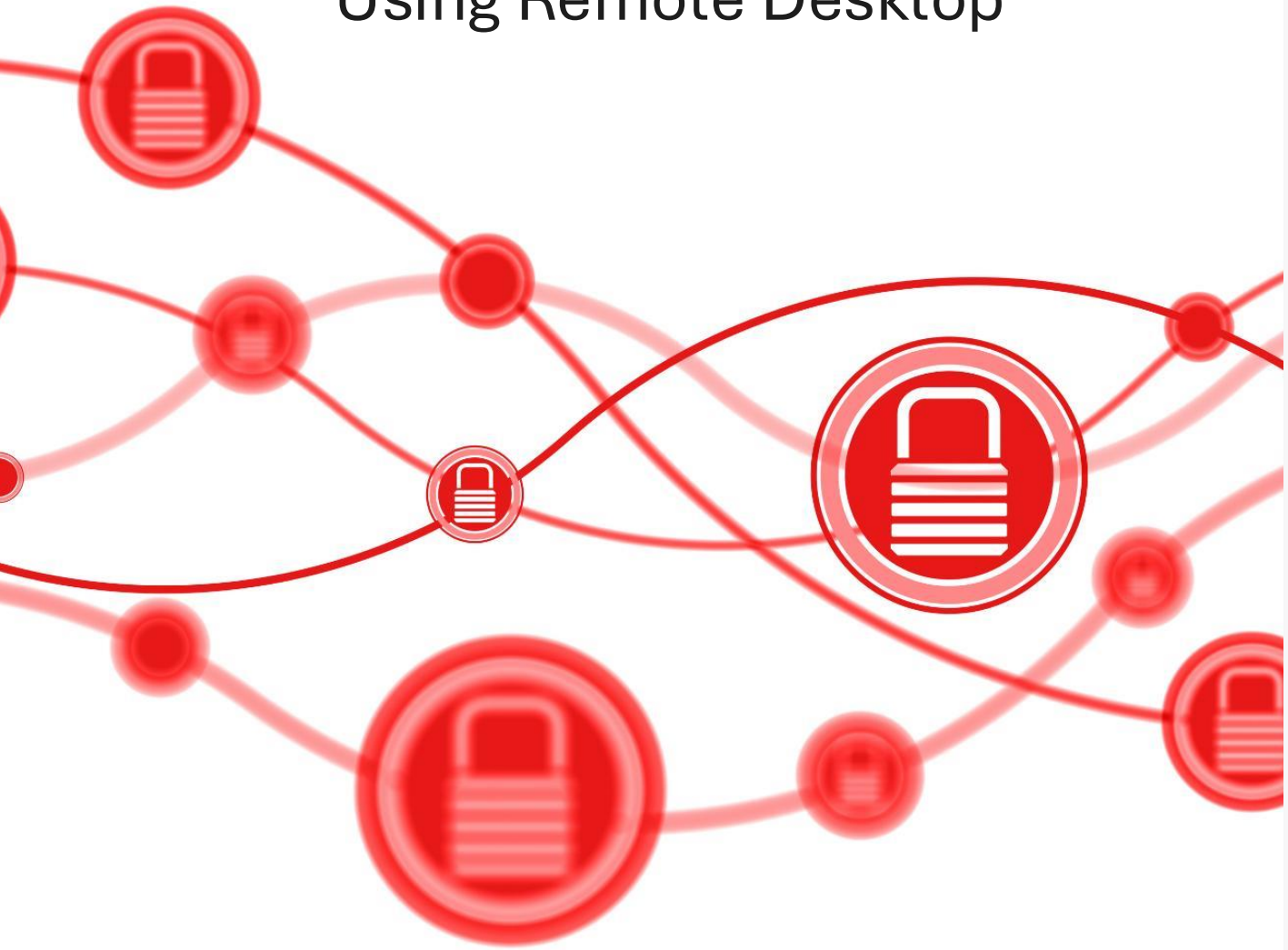
## Selecting and Connecting

Choose the desired network and enter the password to establish a secure connection.

## Signal Strength and Status

Signal strength and connection status indicators help users select the most reliable network.

# Using Remote Desktop



## Remote Desktop Overview

Remote Desktop lets users connect to another Windows 11 computer from any location securely and efficiently.

## Enabling Remote Desktop

To use Remote Desktop, it must be enabled on the host computer through the System settings in Windows 11.

## Applications of Remote Desktop

Ideal for remote work, IT support, and accessing files or applications on a different computer.

# Sharing Resources on the Network

# Managing Shared Folders



## Access Control Management

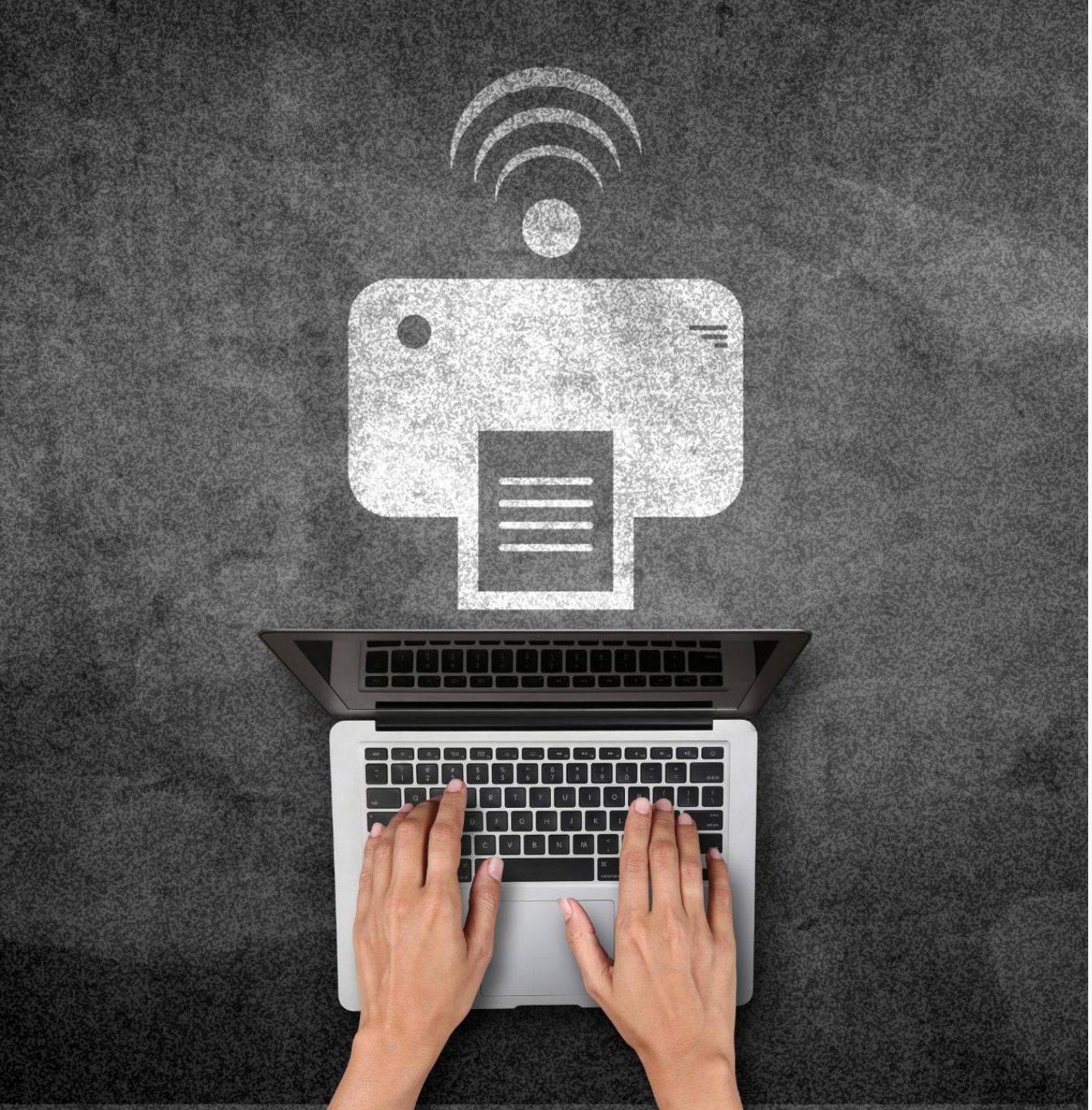
Computer Management console allows detailed control of folder access and user permissions.

## Configuring Shared Folders

Users can create and configure new shared folders for collaboration and resource sharing.

## Security and Collaboration

Proper permission settings ensure data security and facilitate effective teamwork or household sharing.



# Sharing Printers Over the Network

## Enable Printer Sharing

Printer Sharing must be enabled in advanced settings to allow network devices to access the printer.

## Network Printer Connection

Printers must be connected to the network and have correct drivers installed for seamless sharing.

## Benefits of Sharing

Sharing printers reduces hardware costs and improves accessibility for all users on the network.

# Troubleshooting and Diagnostics

# Using Network Troubleshooter



## Purpose of Network Troubleshooter

The tool automatically diagnoses and resolves common network issues for seamless connectivity.

## Accessing the Tool

Users can find the troubleshooter in Settings under Network & Internet for easy access.

## Common Fixes Performed

Includes resetting adapters, verifying connections, and resolving dropped or limited access issues.



# Testing Connectivity with Ping

## Purpose of Ping

Ping tests connectivity by sending packets and measuring response times between devices.

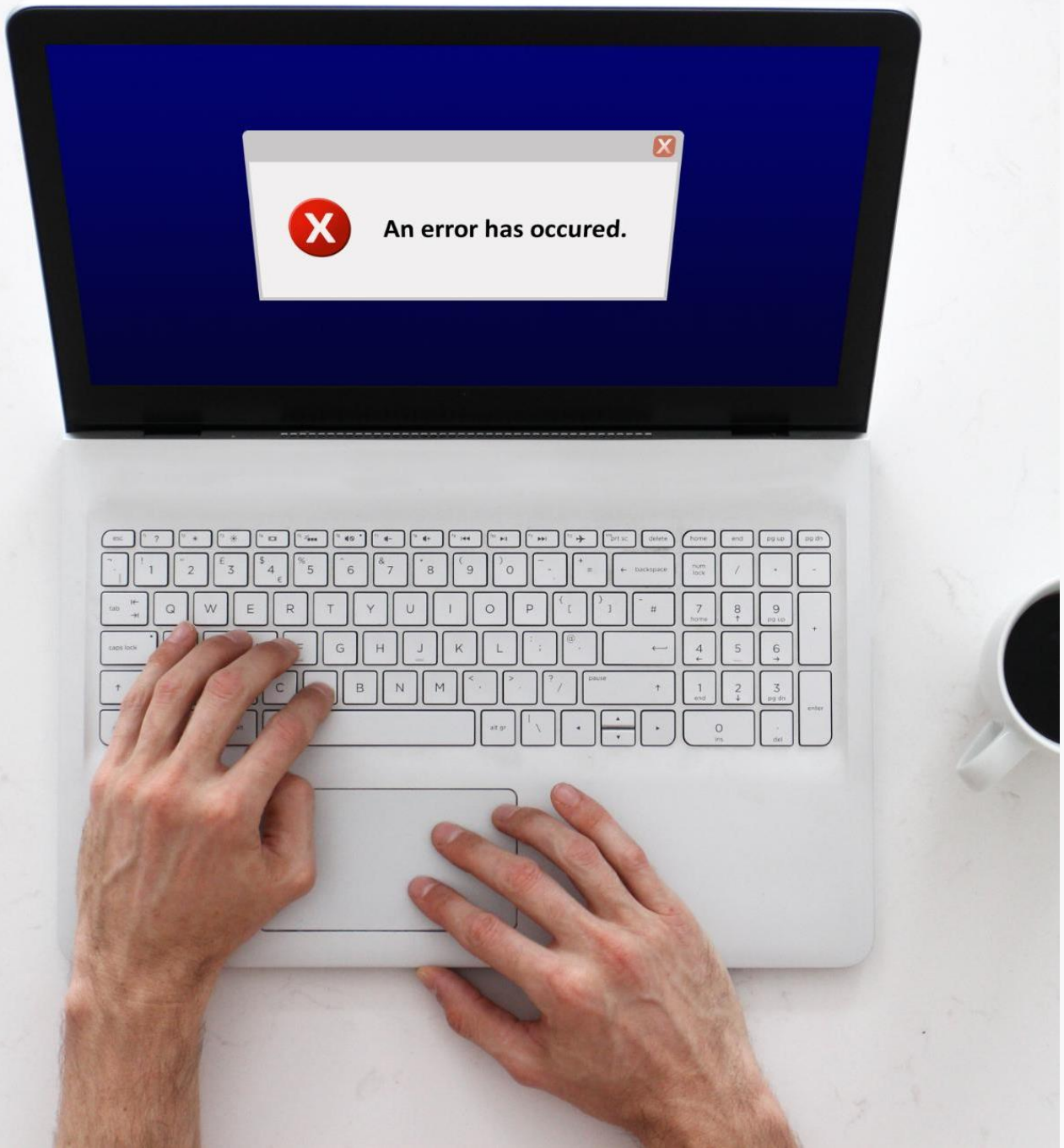
## Identifying Network Issues

Ping helps detect latency problems and unreachable hosts in a network setup.

## Usage in Command Prompt

Ping commands are executed in the Command Prompt to provide immediate connectivity feedback.

# Security and Permissions




# Firewall Configuration for Network Access

## Firewall App Permissions

Windows 11 allows users to specify which apps can bypass firewall restrictions to communicate over the network.

## Security and Functionality Balance

Configuring firewall settings properly ensures security while allowing essential apps like games and remote tools to work.

## Network Communication Enablement

'Allow an app through firewall' setting enables multiplayer games, remote desktop, and file sharing applications.



# Understanding Permissions for Shared Folders

## Permission Types

NTFS and share-level permissions control read and write access to shared folders for users and groups.

## Configuring Permissions

Permissions are set through folder Properties under the Security tab to manage user access rights effectively.

## Importance of Proper Permissions

Correct permissions ensure data integrity and privacy by restricting unauthorized access and avoiding collaboration issues.

# Advanced Network Features




# Using Network and Sharing Center

## Centralized Network Management

Network and Sharing Center offers a unified interface for viewing and managing all network connections and settings easily.

## Adapter and Sharing Configuration

Users can change adapter settings and configure sharing options to customize network behavior and resource access.

## Troubleshooting and Profiles

The tool helps troubleshoot network issues and manage different network profiles for seamless connectivity.




# Prioritizing Network Connections

## Network Priority Settings

Users can prioritize network connections via 'Change adapter options' in Windows 11 to choose preferred networks.

## Adjusting Metric Values

Metric values determine which network adapter has higher priority when multiple connections are active.

## Optimizing Network Performance

Prioritizing connections improves speed and reliability, especially in mixed wired and wireless environments.

# Monitoring Network Usage

# Real-Time Usage in Task Manager




## Real-Time Network Monitoring

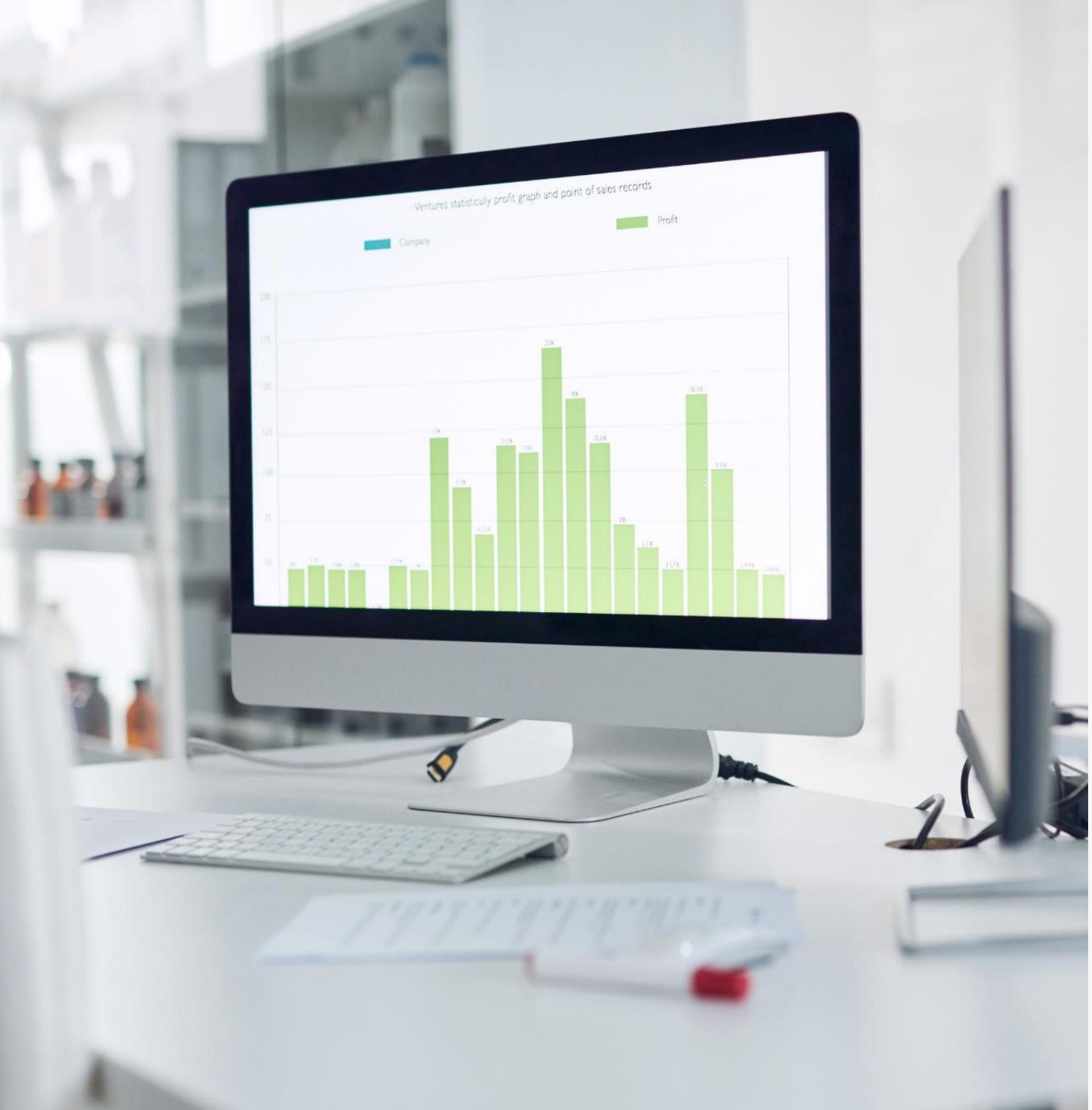
Task Manager displays live network usage data per application for effective monitoring and troubleshooting.

## Bandwidth Consumption Insights

Users can identify apps with high bandwidth usage to optimize resource allocation and prevent slow connections.

## Resource and Security Management

Monitoring network usage aids in managing system resources and detecting unauthorized network activity.




# App History and Network Impact

## Tracking Resource Usage

App History tracks CPU, memory, and network usage over time for system performance analysis.

## Understanding Network Impact

Monitoring network activity helps users manage bandwidth consumption and optimize connectivity.

## Informed Decision Making

Historical app data enables identifying trends and improving app efficiency and data management.

# Command Line Tools for Networking



# Using IPConfig for Configuration Details

## IP Configuration Details

The ipconfig command displays IP address, subnet mask, and default gateway information essential for network setup.

## Network Troubleshooting

Using ipconfig helps diagnose and fix network connectivity problems quickly and efficiently.

## Command Prompt Usage

The ipconfig command is executed in the Command Prompt, providing quick access to vital network data.



# Benefits of Domain Network Profiles

**Centralized Management**

Domain profiles enable centralized control of network settings and security policies across all devices in an organization.

**Group Policy Enforcement**

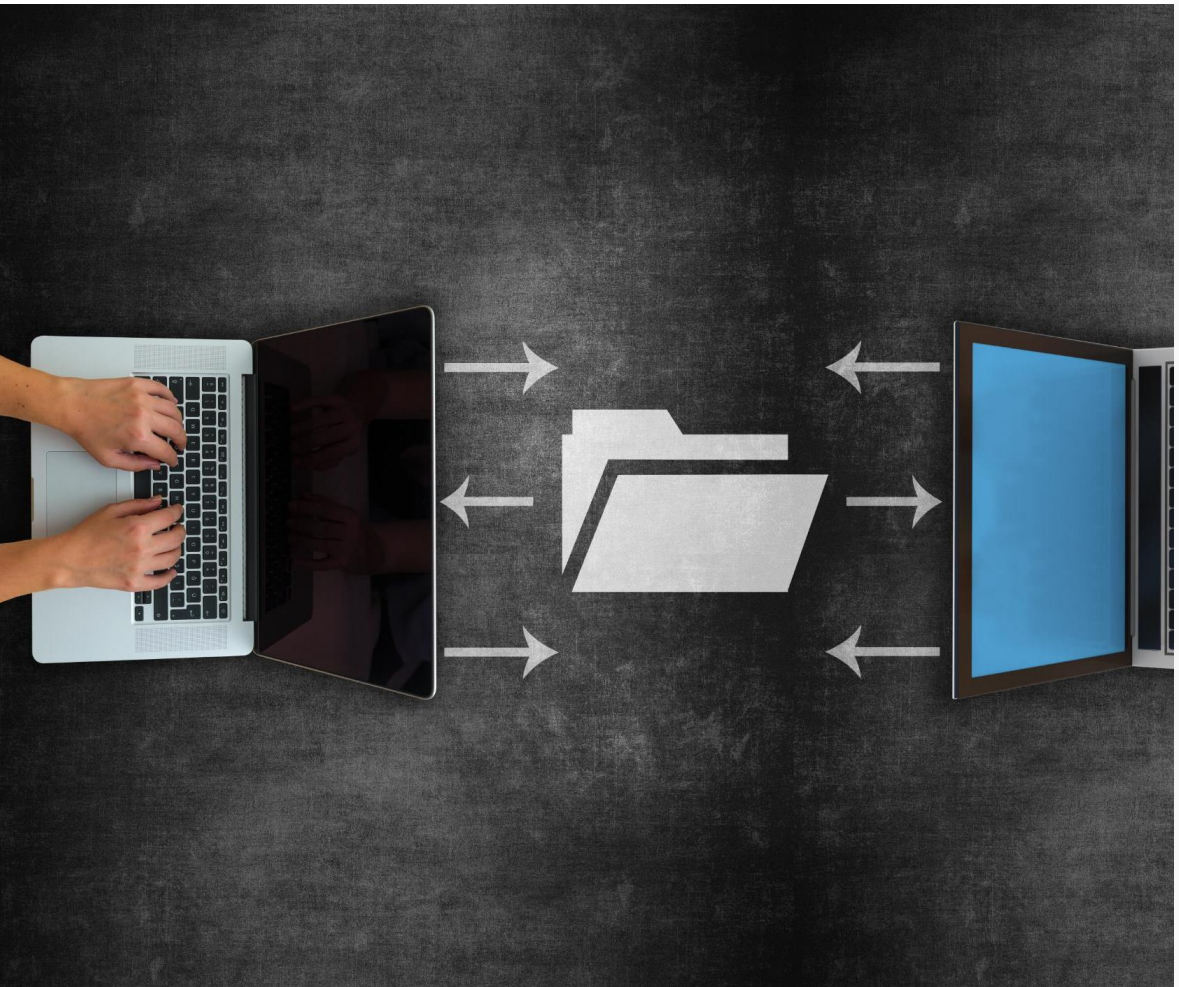
Administrators can enforce consistent group policies to regulate user access and security standards efficiently.

**Enhanced Security and Compliance**

Using domain profiles helps maintain security compliance and reduces administrative overhead in enterprise IT environments.

# Troubleshooting Shared Resources

# Using Shared Folder Troubleshooter



## Issue Identification

The troubleshooter detects permission errors, visibility problems, and network configuration issues affecting shared folders.

## Accessing the Tool

Users can open the Shared Folder Troubleshooter through Windows Settings or the Control Panel for easy access.

## Automated Solutions

The tool provides automated fixes and step-by-step guidance to restore shared folder functionality quickly.

## Collaboration and Access

Maintaining shared folder access ensures seamless collaboration and efficient resource sharing within networks.



# Common Issues with Shared Devices

## Network Discovery Disabled

Network Discovery being turned off blocks devices from detecting each other on the network, causing visibility issues.

## Permissions and Firewall

Proper permissions and firewall configurations are essential to allow shared device visibility across the network.

## Regular Maintenance

Regularly checking and updating network settings ensures a functional and secure environment for shared devices.

# Conclusion and Best Practices

# Summary of Key Networking Concepts



**Network Discovery**

Enabling Network Discovery allows devices to find and communicate with each other on the network effectively.

**Firewall Configuration**

Configuring firewall settings helps protect the network by controlling incoming and outgoing traffic securely.

**Shared Resource Management**

Managing shared resources enables efficient file and device sharing among users within a network.

**Diagnostic Tools**

Using tools like ping and ipconfig helps diagnose and troubleshoot network connectivity issues quickly.

# Tips for Maintaining Network Health



## Regular Permissions Review

Regularly reviewing user permissions helps maintain network security and prevent unauthorized access.

## Update Firewall Rules

Keeping firewall rules updated ensures protection against new security threats and controls network traffic.

## Monitor Network Usage

Using Task Manager to monitor network usage helps identify unusual activity and manage performance.

## Utilize Troubleshooting Tools

Employing built-in troubleshooting tools resolves issues quickly and maintains seamless device communication.